PROVISIONAL PATENT APPLICATION

Angular-Momentum Coherence Energy Storage Cells, Arrays, and Solid-State Power Sources; Weight-Trimmed Potential-Energy Storage Plants; Weight-Trim Vibration-Isolation Platforms; and Machinery Bearing-Load Trim Systems, and Methods of Operating Them

Inventor: Christopher F. Milam

Filing type: U.S. Provisional Application under 35 U.S.C. §111(b)

This specification is complete and self-contained. Applicant's companion applications on support-force control apparatus, media and arrays, and instruments are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Energy conservation is stated in every embodiment; no over-unity operation is claimed or possible. Every example under PROPHETIC EXAMPLES is prospective; no experiment, nonzero support-force response, storage interval, efficiency, or qualified unit is represented as achieved, and no response, sign, magnitude, additivity, retained state, or scaling law is presumed.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media (companion drafts: support-force control apparatus and calibrated modules; transportation and lift systems; measurement instruments; media and array architectures; information and winding-register devices; time-rate metrology; and completion-scheduled computation). Each companion states its own relationship to this disclosure and remains self-contained as to claim-essential matter.

FIELD

Energy storage and conversion; vibration isolation; infrastructure machinery. More particularly: (i) **coherence storage cells**, arrays, and solid-state power sources that store energy in coherently driven angular-momentum populations of long-coherence media and discharge it inductively, together with their charge/store/discharge operating method; (ii) **potential-energy storage plants** whose mass buffer carries calibrated weight-trim modules so that a commanded, measured fraction of the lift-stroke support force is carried through the modules' drive channel, together with the plant's scheduling and metering method; (iii) **weight-trim vibration-isolation and precision-stage platforms** whose compliant supports are selected for the untrimmed fraction of the platform weight; and (iv) **machinery bearing-load trim**

systems applying static and shaft-synchronous support-force offsets to a machine element in situ. Each product is disclosed together with its method of operation, its factory qualification producing the stored empirical map it operates from, and the validity states that govern what a deployed unit may command.

BACKGROUND

Flywheels store kinetic energy in bulk rotation and are bounded by tensile fracture; batteries by electrochemistry; pumped hydro and gravity storage by geography and by the requirement that recovered work equal lift work less losses. Precision industry separately lacks reduced-weight platforms: isolation systems counteract vibration but cannot reduce the static weight that sets suspension stiffness and resonance floors. Rotating machinery is balanced by adding or removing mass, by active magnetic bearings, or by active balancing rings; none trims the static bearing load of the rotor itself.

This disclosure provides two levers, each with an expressly identified accounting lane. (a) A driven, aligned angular-momentum population stores energy as a collective coherent precession state without bulk rotation—a store with no rim speed, discharged inductively from the relaxing population. For the conventional YIG reference species, the measurable alignment (Zeeman) energy $E\_ref \approx N \cdot h\nu \approx 2M\_s \cdot B \cdot V$, of order one joule per hundred grams at 5.6 GHz (D1 below), supplies a conservative reference bound. That species-specific bound is not imposed as a universal ceiling on the separately disclosed POAMS-native winding state or long-storage channel. (b) The support force (measured weight) of a body rigidly borne with a **qualified weight-trim module** may be commanded from the module's stored, empirically established response map. Where such a module class exists, the lift stroke of a potential-energy plant may be partitioned between the hoist channel and the module drive channel, the compliant supports of an isolation platform may be selected for the untrimmed fraction of its weight, and the bearing load of a machine element may be trimmed in situ. Whether any module class becomes qualified is decided by measurement, not presumed; every weight-trim embodiment below is conditional on a qualified module class whose map covers the commanded operating point, and every energy statement below is a ledger that admits no over-unity term.

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, https://cfmilam.github.io/nr-poams-exhibits/). Those pages are applicant's own prior publications and are cited as nonessential background teaching only: no material essential to enablement, written description, or the claims is incorporated from them, and this specification is complete and self-contained.

SUMMARY

The disclosure is organized in three tiers (FIG. 5). The products and their deployed operating methods lead; factory qualification establishes the stored map each product runs from; validity states, recalibration, and safe-off govern what a deployed unit may command.

(1) D1 — Coherence storage cell (600), array (668), and solid-state power source. A long-coherence insulating crystal medium (610) under saturation alignment (620) in a resonant coupling structure (615), charged by pulsed resonant drive (625) into a collective coherent precession state, state of charge read by absorption spectroscopy (635), held with the discharge coupler decoupled (632), discharged through an overcoupled inductive coupler (630) into a tuned load (645) and rectifier (647); a cell controller (640) and cell calibration memory (642) holding the cell's measured energy-relaxation time, stored-energy ceiling, state-of-charge calibration, and validity envelope; arrays with per-cell charge management (650), a common bus (660), an array controller (665), and a common thermal plant (655). Storage for an interval longer than the medium's energy-relaxation time is an express hypothesis with a disclosed identification protocol and acceptance gate; it is not presumed.

(2) Weight-trim module (700) — the common species for D2–D4. A body of ordered medium (702) on a support interface (710); an alignment-field source (704); a coupling structure (706) and locked drive chain (708) at the body's precession resonance; a reaction frame (722) on a separate load path; a calibration memory (712) holding an empirically established **support-force response map** with at least one independently validated nonzero operating point, its uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters, together with the module's measured force-per-drive and hold-energy characteristics; runtime witnesses (716); a setpoint controller (714); safe-off (718). Reproduced in full below; no companion disclosure is relied on.

(3) D2 — Weight-trimmed potential-energy storage plant (670). A mass buffer (672) raised and lowered between levels (680, 682) by hoist machinery (674) with weight-trim modules (676) on the buffer commanded by a scheduler (678) from their maps during selected strokes; energy and momentum metering (679) of every channel; explicit machine sizing for the stroke each machine actually loads; an energy and momentum ledger (FIG. 7, FIG. 8) stated as a table. The plant's advantage is a measured partition of lift-stroke force and energy between the hoist and module channels, reduced lift-stroke hoist and cable loading, distributed actuation, and siting freedom; rate shaping beyond a conventional plant is conditional on the module's measured energy characteristic and is not presumed.

(4) D3 — Weight-trim vibration-isolation platform (690). Platform on compliant supports (692) whose stiffness is selected for the untrimmed fraction $(1-f)$ of platform weight at a design static deflection; modules (694) nulling the commanded fraction f; an isolation controller (696) whose error signal is defined in this document; travel stops (697) rated for full weight; the conditional design rule $\omega' \approx \omega_0\sqrt{(1-f)}$.

(5) D4 — Machinery bearing-load trim system (800). Modules (810) on a machine housing or element commanding a static support-force offset reducing mean bearing load and, within the map's bandwidth, a shaft-synchronous support-axis force offset phase-locked to a once-per-rev reference (808), read against bearing-load sensors (806).

Layered scope (FIG. 5). Every quantity is disclosed at three layers: (L0) the raw synchronous observable —recovered energy or support-force change synchronized with a commanded drive state, always reportable; (L1) the validated map—map-corrected quantity with uncertainty, reportable inside the validity envelope; (L2) controlled operation—cell charge/discharge scheduling, plant scheduling, isolation control, or synchronous offsets commanded from the map. No layer presumes sign, magnitude, additivity, retained state, or scaling; every map admits zero.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 — Coherence storage cell (600): medium (610) in coupling structure (615) under alignment bias (620); charge coupler (625); discharge coupler (630) with decoupling element (632); absorption state-of-charge probe (635); temperature witness (637); controller (640); cell calibration memory (642); tuned load (645); rectifier (647). FIG. 2 — Charge/store/discharge cycle timing: charge pulse train, absorption state-of-charge curve, discharge coupler decoupled during store, inductive discharge current; recovered energy less than supplied. FIG. 3 — Storage array (668): cells on a common charge/discharge bus (660) with per-cell charge management (650), array controller and power interface (665), common thermal plant (655). FIG. 4 — Weight-trim module (700): ordered medium (702) and holder (703); alignment-field source (704); coupling structure (706) and drive chain (708); support interface (710) and optional support-path force sensor (720); calibration memory (712) with stored response map; setpoint controller (714); runtime witnesses (716); safe-off (718); reaction frame (722) and reaction-force witness (724); power and data interface (726). FIG. 5 — Three-tier architecture and layered scope: Tier 2 factory qualification with gates Q1–Q9 producing the stored map → classification as qualified responsive module, qualified cell class, or null reference → Tier 1 deployed operation from the map (L0 → L1 → L2) → Tier 3 validity states, zero checks, recalibration, safe-off. FIG. 6 — Weight-trimmed potential-energy plant (670): shaft or tower with guide structure (677) and guide reaction witnesses; mass buffer (672); motor-generator (674) and optional separate lift motor (675); weight-trim modules (676); scheduler (678); energy and momentum metering (679); upper and lower levels (680, 682); cable-tension sensor (683); grid interface (684); optional guide-reacting lift-share actuator (685). FIG. 7 — Plant energy and momentum ledger: off-peak module drive energy in (686), hoist electrical input work (687), delivered energy (689), losses (688), with the conservation bound; momentum account showing hoist force, module support-force change, measured reaction and guide loads. FIG. 8 — Ledger table for the P2 sizing example (1000 t, 100 m, $f = 0.3$): energy rows, force rows, untrimmed reference column, and module-efficiency columns. FIG. 9 — Weight-trim isolation platform (690): payload (691), platform, compliant supports (692) selected for the untrimmed fraction, modules (694), support-path load cells (693), inertial sensor (695), isolation controller (696), travel stops (697), base (698); and transmissibility plot with the conditional rule $\omega' = \omega_0\sqrt{(1-f)}$ at $k' = (1-f)k$. FIG. 10 — Machinery bearing-load trim system (800): machine (800) with rotating element (802), bearings (804), bearing-load sensors (806), once-per-rev phase reference (808), modules (810), trim controller (812), synchronous support-axis offset (814); phase-scan verification inset.

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels in quotation marks or introduced by "extant label:" identify measured phenomena by their customary names; their use commits neither the specification nor the claims to any entity ontology historically associated with those names. Every measured claim quantity is fixed by a recited instrument reading. Hypothesis parameters are identified expressly.

1. **Internal circulation / internal angular-momentum population** — the ensemble of constituent angular momenta of a medium; the quantity the extant literature labels "electron spin" and measures in aggregate as magnetization M (A/m).
2. **Alignment field (B)** — the static bias field (tesla) establishing a measured alignment state of the population; "saturation alignment" is the measured saturation value; "sense" is field polarity.
3. **Precession resonance** — the absorption resonance of the aligned population under the alignment field (extant label: ferromagnetic resonance), located operationally as the absorption extremum to which the drive servo locks; for a sphere, turn rate $\nu \approx (\gamma/2\pi)\cdot B$ with $\gamma/2\pi \approx 28$ GHz/T, so 5.6 GHz corresponds to $B \approx 0.20$ T.
4. **Oscillating drive** — a time-periodic influence applied through a coupling structure at a stated turn rate, specified by frequency, delivered power, and pulse timing at the stated port (extant labels: microwave or radio-frequency drive).
5. **Turn rate (ν); quanta rate ($\dot{N}$)** — frequency in hertz, with $E = h\nu$; delivered power expressed as a count rate $\dot{N} = P/h\nu$. Each quantum at ν carries energy $h\nu$ and one $\hbar$ of angular momentum.
6. **Collective coherent precession state ("wound state")** — the state of the aligned population precessing coherently at cone angle θ under resonant drive; its stored energy relative to the aligned ground state is $E(\theta) = M_s\cdot B\cdot V\cdot(1 - \cos\theta)$, and its quanta count is $E/h\nu$.
7. *Population coherence time (T_2)** — the decay time of coherent precession, from linewidth or time-domain pulsed response; distinguished from the coupling-structure energy-decay time and from the energy-relaxation time.
8. **Energy-relaxation time (T_1)** — the decay time of stored precession energy of the population to the lattice with the discharge coupler decoupled; the storage interval of a cell is bounded by T_1 unless a longer storage channel is identified under Definition 13.
9. **Coupling-structure energy-decay time (τ_c)** — $Q_L/(2\pi\nu)$ of the loaded coupling structure, measured by ring-down; not identified with T_1 or T_2*.

10. **Sufficient phase coherence** — $\tau \cdot \nu \geq 1$, the population retaining coherence for at least one turn of drive phase; preferred media exceed this by two to three orders of magnitude.
11. **Reference-species conventional storage bound (E_ref)** — for the conventional uniform-precession/magnetization species, $E_{ref} = N_{spin} \cdot h\nu \approx 2M_s \cdot B \cdot V$ for full reversal of the aligned population, with an accessible bound $E(\theta_{max})$ at the largest stable cone angle. This bounds that implementation channel only. It is not asserted as a universal bound on a distinct POAMS whole-turn winding or long-storage channel, whose capacity is measured separately and remains prophetic until identified.
12. **State of charge (SoC)** — the stored energy relative to E_{max}, read as the absorption line shift and saturation depth of a weak probe and corrected for temperature by the stored calibration curve; cross-checked against integrated recovered energy.
13. **Long-storage channel (hypothesis)** — any storage channel whose recovered energy after a hold interval $t_h \gg T_1$ exceeds the prediction from τ_c and T_1; identified only under the protocol and gate of the D1 qualification section, never presumed.
14. **Support force / measured weight** — the force read by a sensor in the support path of a body (newton); a change in support force is the L0 observable of every weight-trim embodiment.
15. **Weight-trim module (700)** — the apparatus of FIG. 4: a body of ordered medium, an alignment-field source, a coupling structure and locked drive at the body's precession resonance, a support interface, a reaction frame on a separate load path, runtime witnesses, a calibration memory holding a stored support-force response map, a setpoint controller, and safe-off; reproduced in full in this specification.
16. **Stored support-force response map** — the data structure held in a module's calibration memory: the change in support force relative to the drive-off baseline at each validated drive state, with uncertainty or covariance, hysteresis, time constants, bandwidth, thermal and orientation coefficients, the force-per-drive characteristic (support-force change per unit absorbed drive power), the hold-energy characteristic (drive energy per unit time to hold a stated support-force change), the identity of every independently validated operating point, and the provenance record, validity envelope, runtime limits, and safe-off parameters of Definitions 17–19. A map may contain only null points.
17. **Provenance record** — unit or class identity; qualification apparatus and force references; date; raw-data identifier; fit method; replication record; gates passed per operating point.
18. **Validity envelope; permissible interpolation region; runtime limits** — the region of drive-state and environmental variables over which the map was validated, the region within which interpolation between validated points is permitted, and the runtime limits (temperature, orientation, force-axis field gradient, vibration, absorbed power and duty, bandwidth and slew, drift, zero-check interval) outside which the controller reports out-of-envelope. Extrapolation is not permitted.
19. **Safe-off** — the stored state on validity failure, fault, or command: oscillating drive off; alignment field to a stored safe magnitude and sense (or unchanged in a fixed-bias unit); last valid record

preserved; recalibration flag set. In a weight-trim embodiment safe-off returns the full untrimmed weight to the mechanical support path within the stated travel or rating limits.

20. **Qualified responsive module; qualified module class; null reference** — a module (or class represented by qualified units) whose map contains at least one independently validated nonzero operating point with uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters is a *qualified responsive module* (*class*). A module whose map contains only null points is a *null reference*: it accepts test-state commands and reports L0 and L1 but has no enabled setpoint mode. Neither status presumes response, sign, magnitude, additivity, or scaling beyond validated points.
21. **Qualified cell class** — a storage cell design whose cell calibration memory holds a measured energy-relaxation time, stored-energy ceiling, SoC calibration curve, discharge coupling ratio, leakage isolation, and validity envelope established under the D1 qualification section; a *long-storage-qualified* cell class additionally holds an identified long-storage channel under Definition 13.
22. **Support interface (710); reaction frame (722)** — the mechanical path through which the body's support force is transmitted to the supported structure (buffer, platform, machine element, or force sensor); and the frame carrying the alignment-field source and coupling structure whose loads pass through a separate reaction load path. A commanded support-force change accounted for within tolerance by an equal and opposite change on the reaction path or on a guide structure is an artifact and is reported invalid.
23. **Trim setpoint (f)** — a commanded change in support force expressed as a fraction f of the drive-off baseline support force of the body or of a load rigidly borne with it, or as a time-varying profile within the map's bandwidth and slew limits; the achieved fraction is a reported, not presumed, quantity.
24. **Runtime witnesses** — body and mount temperature, orientation, a force-axis field-gradient witness, acceleration and vibration, incident, reflected, and absorbed drive power and duty, lock state, and, where present, support-path and reaction-path force.
25. **Validity states** — *valid* (setpoint permitted), *degraded* (L0/L1 reported, setpoint held or reduced), *out-of-envelope* (setpoint inhibited), *invalid* (map or gate failure; recalibration flagged), *safe-off* (Definition 19).
26. **Array allocation map** — for a plurality of modules bearing a common load, the measured relation between per-module drive states and the change in system-level support force, established under sequential, in-phase, phase-scrambled, and inactive-module states; the sum of individual responses is not presumed.
27. **Energy and momentum ledger** — the metered account of a plant, platform, or machine: hoist or machine electrical input, module drive electrical input, generator or load output, thermal and friction losses, stored potential, and the support-path, reaction-path, cable, and guide forces; every embodiment satisfies $E_{out} \leq E_{in}(modules) + W_{machine} - Q_{losses}$ and reports round-trip efficiency as a measured quantity less than one.

28. **Lift-share actuator (685)** — a conventional alternative or auxiliary actuator on the buffer reacting against the guide structure (linear motor, rack, or friction drive); it provides an enabled conventional species of the two-channel lift partition and is distinguished from a weight-trim module, whose support-force change is not a guide reaction.

THE WEIGHT-TRIM MODULE (700) — COMMON SPECIES FOR D2–D4 (FIG. 4)

Body of ordered medium (702) and holder (703). The reference body is a polished single-crystal yttrium-iron-garnet (YIG) sphere, diameter 2–15 mm (reference 10 mm; volume 0.5236 cm^3; mass approximately 2.71 g at 5.17 g/cm^3; drive-off baseline support force approximately 26.6 mN). Infrastructure-scale modules carry a plurality of such bodies, larger boules, or tiled assemblies, each qualified on its own measurements; other garnets, ferrites, and engineered media of sufficient phase coherence are candidate media qualified the same way. The holder is low-loss and nonmagnetic (quartz, sapphire, PTFE, alumina) and forms the upper element of the support interface.

Support interface (710) and optional support-path force sensor (720). The holder transmits the body's support force rigidly to the supported structure—the mass buffer (D2), the isolation platform (D3), or the machine housing or element (D4)—so that a change in the body's support force appears as a change in the support force of the rigid assembly by ordinary statics. The map used in a load-bearing configuration is established or confirmed with the module in its mounting class and orientation; a bench offset is not presumed to transfer unchanged. Where a support-path force sensor (720) is present it supplies closed-loop feedback and runtime zero checks; otherwise the module operates feedforward from the map with witness monitoring and interleaved drive-off zero checks.

Alignment-field source (704). A permanent-magnet yoke with trim winding, an electromagnet, or a coil-and-yoke assembly producing a static field of commanded or fixed magnitude and sense, reference 0.15–0.25 T for the reference body. The sign of support-force change associated with either sense is taken from the map, never from the field polarity.

Coupling structure (706) and drive chain (708). A TE-mode rectangular cavity (reference TE_{102} near 5.6 GHz), loop-gap resonator, re-entrant resonator, waveguide, or coil places the body at a mapped transverse magnetic-field antinode of the drive with the drive's magnetic axis orthogonal to the alignment field. The drive chain comprises a voltage-controlled source spanning the resonance (reference 4–8 GHz), a pulse and burst modulator with a controller-enforced duty bound, a power amplifier sized to the map, a directional coupler, and incident, reflected, and transmitted power detectors. The controller servo-locks to the located precession resonance and records mode, linewidth, loaded quality factor, and ring-up and ring-down times. Pulses are nonoverlapping; higher delivered energy is implemented as bursts separated by recovery intervals or as thermally matched continuous-wave operation.

Reaction frame (722) and reaction-force witness (724). The alignment-field source, coupling structure, feeds, thermal hardware, and shields are carried by a frame mechanically independent of the support interface, their loads passing through a reaction load path to the supporting structure or base. A reaction-force sensor on this path is a runtime witness: a commanded change in support force accounted for within tolerance by an equal and opposite change on the reaction frame is an actuator artifact and the module reports invalid.

Calibration memory (712). Holds the stored support-force response map (Definition 16), provenance record, validity envelope and permissible interpolation region, runtime limits, safe-off parameters, unit or class identifier, validity-state history, zero-check history, and recalibration schedule. The controller refuses a map whose provenance does not match the unit or class identifier.

Setpoint controller (714), runtime witnesses (716), safe-off (718), interface (726). The controller receives a trim setpoint (Definition 23) from the plant scheduler, isolation controller, or trim controller through the data interface (726); verifies that the setpoint lies within the permissible interpolation region and inhibits it otherwise; inverts or interpolates the map to the drive state; commands the alignment-field source and drive chain within stored slew and bandwidth limits; regulates by feedforward from the map and, where a support-path force sensor is present, by feedback with loop gains bounded by the map's bandwidth, hysteresis, and time constants; compares the runtime witnesses (Definition 24) with the runtime limits on every control cycle; reports the achieved change with propagated uncertainty and validity state; and executes safe-off (Definition 19). Setpoint mode is enabled only for a qualified responsive module; a null reference executes test-state commands only. The power interface carries drive and field power and is metered.

Momentum and energy account of a module. The module's L0 observable is the change in support force at the support interface synchronized with the commanded drive state. Its reaction path and, in D2, the guide structure are separately instrumented; a change accounted for by either is rejected as an artifact. Within the applicant's hypothesis a validated change is a change in the body's constraint bill against the coupled circulation it participates in, not a thrust against any local structure; that reading is an interpretation, and the plant, platform, and machine embodiments below operate from the measured change alone. The module's drive energy is metered at the power interface; the hold-energy characteristic of the map (drive energy per unit time per unit support-force change) is a measured quantity entering every ledger, and no value of it is presumed.

THREE-TIER ARCHITECTURE, LAYERED SCOPE, AND QUALIFICATION (FIG. 5)

Tier 1 — products and deployed operating methods (leading). The cell (600) and array (668) of D1, the plant (670) of D2, the platform (690) of D3, and the machine system (800) of D4, each operated by an ordinary operator from its stored map by the deployed operating steps of its section; the operator does not perform the qualification laboratory.

Tier 2 — factory qualification producing the map. *Weight-trim modules:* the module body on a sample force sensor mechanically separate from the coupling structure, alignment-field source, feed lines, and drive hardware; randomized and optionally blinded resonant-drive, detuned absorbed-energy-matched, drive-off, field-off, reference-material, position-swap, and sham states in thermally settled field-sense blocks; synchronized logging of support force, reaction force, drive power, temperature, pressure, vibration, acceleration, strain, field, and force-axis gradient; gates **Q1** repeatability across blocks and days, **Q2** resonance contrast against the absorbed-energy-matched detuned state, **Q3** material contrast against matched reference bodies and swaps, **Q4** magnetic-force exclusion (mapped force-axis component of $\nabla(\mu \cdot B)$ acting on the measured moment cannot account for the response; residual gradient uncertainty below a preregistered fraction of the imitating gradient), **Q5** thermal and radiometric exclusion, **Q6** mechanical, electrical, and reaction-force exclusion, **Q7** orientation and sense parity learned from held-out data, **Q8** statistical validity after multiple-comparison correction, **Q9** independent confirmation on a second apparatus, body, run series, or laboratory. The output is a map with uncertainty and an artifact classification—including a valid null—written to the calibration memory with provenance, region, limits, and safe-off parameters; the unit is classified as a qualified responsive module or a null reference. Production-class maps are established from a plurality of qualified units, and an additional unit is accepted under the class map only after an acceptance test at validated operating points against a separate force reference. Array qualification of an assembled plurality produces the array allocation map (Definition 26). *Storage cells:* the D1 qualification section below.

Tier 3 — validation, recalibration, and fail-safe. Validity states (Definition 25); runtime zero and reference checks at the stored interval; drift-triggered envelope shrinkage or recalibration; a service path (reduced re-qualification: drive-off and sense-reversed zero checks plus one validated point against a separate force reference, else return to full qualification); and the rule that a deployed unit never commands an operating point its map does not support.

Layered scope. L0 raw synchronous observable → (gates Q1–Q9 or the cell gates) → L1 validated map → (runtime validity gate) → L2 controlled operation. Every product section below states which layer each output belongs to.

What qualification does not presume. Sign, magnitude, linear scaling with population or power, additivity across modules or cells, retained state after drive termination, storage beyond T_1, and any lift, null-crossing, or hold power are outputs of measurement; each is absent from the map until measured and validated.

DETAILED DESCRIPTION — D1: COHERENCE STORAGE CELL (600), ARRAY (668), AND SOLID-STATE POWER SOURCE (FIGS. 1–3)

Cell architecture (FIG. 1). A storage medium (610)—reference class: single-crystal yttrium iron garnet, $Y_3Fe_5O_{12}$, an insulating ferrimagnet of sufficient phase coherence; a cryogenic variant is described below—is held in a resonant coupling structure (615) under a saturating alignment bias field (620) from a permanent-magnet yoke or electromagnet; for the reference 5.6 GHz Kittel-mode resonance the bias is approximately 0.20 T. A charge coupler (625) delivers pulsed resonant drive on a first coupling port. A discharge coupler (630) on a second port couples the medium to a tuned load (645) through a decoupling element (632)—a switch, detuning element, or variable coupling iris—so that the load is disconnected or detuned during store. A weak continuous-wave absorption probe (635) reads state of charge. A temperature witness (637) on the medium and coupling structure corrects the probe and enforces the thermal envelope. A cell controller (640) manages charge, store, and discharge from a cell calibration memory (642) holding the measured energy-relaxation time T_1, the stored-energy ceiling E_{max}, the SoC calibration curve versus temperature, the discharge coupling ratio, the leakage isolation, and the validity envelope. A rectifier (647) on the load converts the recovered gigahertz output to direct current, the cell thereby constituting a solid-state electrical power source.

Operating method (FIG. 2). *Charge:* pulsed resonant drive at the located precession resonance advances the aligned population into a collective coherent precession state (Definition 6) within coherence-limited windows, the pulse schedule repeated at the pump-and-relax optimum established from the measured T_1 and T_2*; the controller meters absorbed energy and inhibits charge beyond the stored ceiling. *Store:* the bias is held and the discharge coupler is decoupled (632) so that the state is not drained into the load in the coupler's own time constant; the storage interval is the measured T_1 class unless the cell class is long-storage-qualified (Definition 21). *Discharge:* the decoupling element connects the overcoupled discharge coupler, and the relaxing precession induces an output in the pickup structure delivered to the tuned load (645) and rectified (647). *Report:* recovered energy, absorbed drive energy, and round-trip efficiency are metered and reported as measured quantities, bounded below one.

Stored-energy ceiling — design rule. Energy density is bounded by drive input less losses, and more tightly by the alignment energy of the population. For 100 g of YIG (molar mass 737.9 g/mol; 0.1355 mol of formula units; $N_{fu} \approx 8.16\times10^{22}$; net moment 5 μ_B per formula unit at low temperature), full reversal of the aligned population absorbs $N_{max} \approx 5 \cdot N_{fu} \approx 4.1\times10^{23}$ quanta, which at $h\nu(5.6\ \text{GHz}) = 3.71\times10^{-24}$ J is $E_{max} \approx 1.5$ J; the room-temperature cross-check $2M_s \cdot B \cdot V$ with $\mu_0 M_s \approx 0.178$ T, B = 0.20 T, and V = 19.3 cm³ gives ≈ 1.1 J. The volumetric bound is therefore of order 56–78 kJ/m³ per the temperature. The accessible energy at a stable cone angle θ is $E(\theta) = M_s \cdot B \cdot V \cdot (1 - \cos\theta)$; at θ = 3° this is ≈ 1.4×10^{-3} of the full-reversal bound (≈ 1.5 mJ per 100 g), and the largest stable θ for the selected specimen and mode (extant label: Suhl threshold) is a measured quantity stored in the cell calibration memory. The cell's advantages are accordingly solid-state construction, no moving parts, pulse-rate agility, microsecond-class response, and a natively countable state of charge—not volumetric energy density.

Quanta count. Delivered power is a quanta rate, $P = \dot{N} \cdot h\nu$: a 50 W drive at 5.6 GHz delivers $\dot{N} = P/h\nu = 1.35\times10^{25}$ quanta per second, each carrying energy hν and one ħ of angular momentum, so that the 100 g reference cell reaches its full-reversal ceiling in $N_{max}/\dot{N} \approx 30$ ms of continuous drive; every quantum delivered beyond the ceiling thermalizes. State of charge is thus, physically, an accumulated count, tracked by the spectroscopic observable and run back through the discharge coupler; the same E = hν arithmetic underlies the SI realization of the volt (Josephson constant $K_J = 2e/h = 483{,}597.848$ GHz/V), cited here only to anchor the counting, not as a cell mechanism.

Discharge coupling — design rule. The recoverable fraction of stored energy is approximately $\kappa_{ext}/(\kappa_{ext} + \kappa_{int})$, where κ_{ext} is the external coupling rate to the load through the discharge coupler and κ_{int} the intrinsic relaxation rate $1/T_1$; the discharge coupler is therefore overcoupled ($\kappa_{ext} > \kappa_{int}$, preferably $\kappa\_ext \gg \kappa\_int$) and is decoupled during store. Rectification of the gigahertz output (Schottky or synchronous rectification) is ordinary art.

Thermal envelope. 50 W absorbed by a 100 g crystal (specific heat ≈ 0.6 J g^{-1} K^{-1}) heats it at ≈ 0.8 K/s if fully dissipated; the controller enforces pulse duty and the array's common thermal plant (655) removes the heat. The SoC probe is temperature-corrected because drive heating alone lowers M_s and shifts the line.

Cryogenic variant. Operation at or below 77 K is an optional candidate condition: the selected medium's linewidth, T_1, and T_2* are measured at each temperature and stored in the calibration memory; cooling is not presumed to lengthen T_1 (pure YIG narrows modestly at low temperature; rare-earth-contaminated YIG exhibits a low-temperature linewidth peak). The controller extends the store interval only to the measured T_1 at the operating temperature.

Long-storage channel — hypothesis and identification protocol. The applicant hypothesizes that a coherently wound population may retain a fraction of its stored energy in a channel whose decay is slower than T_1. The hypothesis is identified, never presumed, by the following protocol executed at cell-class qualification: (a) measure τ_c (ring-down with the medium detuned off resonance), T_2* (linewidth and pulsed response), and T_1 (energy decay with the coupler decoupled); (b) charge to a stated SoC, hold for $t\_h \gg T_1$ (preferably $t\_h \geq 10 \cdot T_1$), discharge, and meter recovered energy; (c) compare with the prediction from τ_c and T_1; (d) repeat with bias detuned off resonance (same drive, same coupling structure) to separate coupling-structure photons from the population; (e) repeat with the medium removed and with drive-off; (f) gate the discharge measurement after drive-off with isolation of at least 60 dB between charge and discharge ports; (g) correct for temperature-driven line shift. The **acceptance gate** is: recovered energy after t_h exceeds the τ_c–T_1 prediction by more than the preregistered uncertainty, survives (d)–(g), and is independently replicated. Only then is the cell class *long-storage-qualified* and its store interval extended in the calibration memory. Absent that gate, the store interval is T_1 class.

Verification and artifact controls (all cell classes). (i) *Coupling-structure ring-down versus population relaxation:* a Q ≈ 10^4 structure at 5.6 GHz rings down in ≈ 0.3 μs, comparable to YIG's T_1 (≈ 0.3–1 μs at room temperature for Gilbert damping 10^{-4} to 3×10^{-5}); the bias-detuned control of step (d) isolates the population's contribution. (ii) *Thermal line shift versus SoC:* the temperature witness (637) and the zero-drive line-shift-versus- temperature calibration correct the SoC reading, which is cross-checked against integrated recovered energy. (iii) *Drive leakage:* direct feedthrough from charge to discharge port is excluded by gating and by the medium-removed sham. These controls support the device; they are not the claimed product.

Array (668) and power source (FIG. 3). Cells (600) on a common charge/discharge bus (660) with per-cell charge management (650) enforcing each cell's ceiling and thermal envelope, an array controller and power interface (665), and a common thermal plant (655). Rectified output per cell or per string constitutes the solid-state power source. A time-of-use method charges the array during a first period and discharges into a load during a second period selected by a rate or demand criterion; dispatch is limited to metered stored energy less measured losses.

Cell-class qualification (Tier 2 for D1). For each cell class: measure T_1, T_2^*, τ_c, E_{max} (from measured M_s, B, V), the largest stable cone angle, the SoC calibration curve versus temperature, $\kappa_{ext}/\kappa_{int}$, leakage isolation, and round-trip efficiency at stated SoC; run the long-storage protocol above; write the results, validity envelope, and long-storage status to the cell calibration memory (642). A deployed cell schedules charge, store, and discharge only within that record.

DETAILED DESCRIPTION — D2: WEIGHT-TRIMMED POTENTIAL-ENERGY STORAGE PLANT (670) (FIGS. 6–8)

Plant architecture (FIG. 6). A shaft, tower, or slope rail (670) with a guide structure (677) instrumented with guide-reaction witnesses; a mass buffer (672) movable between an upper level (680) and a lower level (682); hoist machinery comprising a motor-generator (674) and, in one embodiment, a separate lift motor (675); a cable-tension sensor (683); a plurality of weight-trim modules (676), each a module (700) of FIG. 4 mounted so that its support interface (710) transmits its body's support force rigidly to the buffer and its reaction frame (722) loads pass to the buffer through a separate, instrumented path; a scheduler and plant controller (678); energy and momentum metering (679) of hoist electrical input, module drive electrical input, generator output, cable tension, module support-path force, module reaction-path force, and guide-reaction force; and a grid interface (684). Where the plurality of modules bears the buffer in common, the scheduler allocates per-module setpoints from a measured array allocation map (Definition 26), not from presumed additivity. Optionally a conventional lift-share actuator (685) reacting against the guide structure provides an auxiliary or alternative lift channel.

Operating method. *Off-peak lift:* the scheduler verifies every module is a qualified responsive module in the valid state whose map covers the commanded fraction f; commands the trim setpoint from the map; the buffer, at measured support force $(1-f)\cdot m\cdot g$, is raised by the hoist while the modules' drive energy is metered from the same off-peak supply (686). *Release or hold at the upper level:* the modules are commanded to safe-off (weight returns to the hoist path within cable and brake ratings) or hold at the setpoint per the descent embodiment selected. *On-peak descent:* the buffer descends through the generator (674) at the scheduled rate. *Metering:* the ledger (Definition 27) is maintained on every stroke; dispatch is limited to metered stored potential less measured losses. *Report:* round-trip efficiency is a measured output.

Force-per-drive characteristic. The support-force change per module and its drive energy cost are taken from each module's stored map (force-per-drive and hold-energy characteristics), never presumed. For the P2 example the modules must together carry $f\cdot m\cdot g = 0.3 \times 9.81$ MN ≈ 2.94 MN for the duration of the lift; the number of modules, their mass fraction riding on the buffer (which enters m in the ledger), and their drive power follow from the class map. The fraction ladder—at least one percent, ten percent, fifty percent, and substantially the whole—is available only where the class map covers that fraction; at substantially the whole the machine would perform substantially no lift work and the plant would operate as a drive-to-generator converter, a prophetic embodiment conditional on module force density that is not represented as achieved.

Machine sizing — explicit. A machine rated for $(1-f)\cdot m\cdot g$ cannot brake a full-weight descent at the lift speed. The disclosure therefore selects one of: **(a) reference:** the generator is rated for the full-weight descent it brakes ($P_{gen} = m\cdot g\cdot v_{descent}$), the lift channel is rated for $(1-f)\cdot m\cdot g\cdot v_{lift}$; where a single motor-generator (674) is used its nameplate rating is set by the larger stroke load—the full-weight descent—and the trim reduces lift-stroke hoist force, cable tension, motor current, and lift-channel thermal load, not the nameplate; **(b)** a separate lift motor (675) rated for $(1-f)\cdot m\cdot g\cdot v_{lift}$ and a generator (674) rated for $m\cdot g\cdot v_{descent}$; **(c)** descent at reduced speed $v_{descent} = (1-f)\cdot v_{lift}$ through a motor-generator rated for $(1-f)\cdot m\cdot g\cdot v_{lift}$, delivered on-peak power being reduced in proportion and stated in the schedule; **(d)** modules held at a fraction f' during descent, the motor-generator rated for $(1-f')\cdot m\cdot g\cdot v$, recovered mechanical energy $(1-f')\cdot m\cdot g\cdot h$, and the modules' descent-stroke drive energy entered as a cost unless the class map records otherwise. Cables, brakes, and buffer structure are rated for full weight in every embodiment because safe-off returns the full weight to the hoist path.

Energy and momentum ledger (FIG. 7). $E_{out} \le E_{in}(\text{module drive}) + W_{hoist} - Q_{losses}$. For the lift stroke the hoist performs $(1-f)\cdot m\cdot g\cdot h$ of mechanical work and the modules draw $f\cdot m\cdot g\cdot h/\eta_d$ of electrical energy, where η_d is the module channel's measured energy characteristic (mechanical lift work transferred per unit module drive energy), an output of qualification and not presumed; the hoist draws $(1-f)\cdot m\cdot g\cdot h/\eta_{hoist}$; descent returns at most $m\cdot g\cdot h\cdot\eta_{gen}$. Total off-peak draw is unchanged from a conventional plant when $\eta_d = \eta_{hoist}$, larger when $\eta_d < \eta_{hoist}$, and smaller only if $\eta_d > \eta_{hoist}$; round-trip efficiency is below one in every case. The **momentum account** records hoist (cable) force, the modules' summed support-force change at their support interfaces, module reaction-path force, and guide-reaction force; a support-force change accounted for by reaction-path or guide force is an artifact and invalidates the stroke's trim record. The buffer's momentum is exchanged with the hoist through the cable and with the guide through measured contact; the modules' change in the buffer's constraint bill is the measured L0 quantity.

Advantage, stated conditionally. The plant partitions lift-stroke force and energy between the hoist channel and the module drive channel by a commanded, measured fraction, reducing lift-stroke hoist force, cable tension, and lift-channel rating, permitting distributed actuation on the buffer, and easing siting (shorter or lighter hoist plant for a given buffer). Rate shaping of grid draw and delivery beyond that of a conventional plant is available only if the module channel's measured η_d exceeds the hoist channel's or if the module drive draw can be time-decoupled from the lift stroke; neither is presumed, and the scheduler commands only schedules the metered ledger supports. The invention is architectural (force partition, sizing, siting), not thermodynamic.

Enabled conventional species. With the lift-share actuator (685) in place of or alongside the modules, the two-channel lift partition, scheduler, sizing options (a)–(d), and ledger are performed identically with the actuator's reaction against the guide structure metered; that species is conventional in mechanism and is disclosed so the plant architecture is enabled irrespective of module qualification.

DETAILED DESCRIPTION — D3: WEIGHT-TRIM VIBRATION-ISOLATION PLATFORM (690) (FIG. 9)

Platform architecture. A metrology platform (690) carrying a payload (691) on compliant supports (692) over a base (698); one or more weight-trim modules (694), each a module (700) whose support interface (710) transmits its body's support force rigidly to the platform; support-path load cells (693) in the compliant-support path; an inertial sensor (695) on the platform; travel stops (697) rated for the full platform weight; and an isolation controller (696).

Conditional design rule. For platform mass m on supports of total stiffness k, the suspension natural frequency is $\omega_o = \sqrt{k/m}$. A constant support-force offset alone changes the static equilibrium position, not k or m, and leaves ω_o unchanged. The platform of this disclosure therefore **selects the compliant supports' stiffness for the untrimmed fraction (1−f) of platform weight at the design static deflection**, $k' = (1-f)\cdot k$, whereupon the suspension natural frequency is $\omega' = \omega_o\cdot\sqrt{1-f}$ relative to supports selected for full weight at the same deflection. Where the module's support-force change is position-dependent it contributes a stiffness k_{mod} (positive or negative) measured at qualification and entered as $\omega' = \sqrt{(k' + k_{mod})/m}$. Inertial mass is untouched by weight trim; the rule is a design prescription conditional on stiffness selection and a qualified module class, not a physical necessity.

Error signal — defined. The isolation controller's trim error is $e_F = F_{meas} - f\cdot W$, where W is the platform's drive-off baseline weight from the load cells (693) and F_{meas} is the module's achieved support-force change read from its support-path force sensor (720) or from the change in the load-cell sum; the module setpoint is regulated on e_F within the map's bandwidth. The isolation loop uses the inertial sensor (695) for velocity feedback (active damping) and, at high f, for positive stiffness augmentation; both loops are gain-bounded by the map's bandwidth, hysteresis, and time constants.

f → 1 requirements. At residual stiffness k′ and travel margin x_m, the trim-force error must satisfy $\delta F < k'\cdot x_m/S$, S a stated safety factor; the loop must supply the stiffness and damping the supports no longer provide; the module's hold-energy characteristic (drive energy per unit time to hold $f\cdot W$) is taken from the map, not presumed; the stops (697) are rated for full weight because on safe-off the full weight returns to supports selected for $(1-f)\cdot W$, which alone would deflect by $x_{sag}/(1-f)$, so safe-off lands the platform on the stops—safe, not isolating. "Passive-safe" is therefore used here only in that sense. At f substantially unity the platform is substantially weight-null, free within its travel limits, its suspension resonance substantially eliminated—an embodiment conditional on a qualified module class whose map covers f, on δF, and on the augmented loop; sub-hertz isolation floors without pneumatic towers are the intended, not achieved, result.

Embodiments. Optical tables, lithography stages, torsion-balance and interferometer test benches, gravitational-wave test benches, spacecraft optical benches. **Controls:** the resonance lowering is generic to any constant-force compensator with re-selected stiffness; identical supports at the same f with a counterweight or magnetic compensator separate the generic effect from any module-specific effect, and k' is verified independently by static load-deflection so the measured ω' is attributed to stiffness.

DETAILED DESCRIPTION — D4: MACHINERY BEARING-LOAD TRIM SYSTEM (800) (FIG. 10)

Architecture. A machine (800) with a rotating or reciprocating element (802) in bearings (804); bearing-load sensors (806) (load cells or accelerometers at the bearing pedestals); a once-per-revolution phase reference (808) (encoder or keyphasor); one or more weight-trim modules (810), each a module (700) mounted rigidly to the bearing housing, pedestal, or a non-rotating machine element so that its support-force change appears in the bearing load by statics, or—where the map has been established in that mounting class—to the rotating element; and a trim controller (812).

Two operating modes. *(i) Static bearing-load trim:* the controller commands a constant trim setpoint reducing the mean support-axis bearing load of the rotor assembly by a commanded, measured fraction—a genuine weight trim of the supported element. *(ii) Synchronous support-axis force offset (814):* rotating imbalance produces a rotating inertial force at shaft frequency, and reciprocating imbalance its harmonics; these are inertial, not weight, forces. Within the map's bandwidth, the controller commands a time-varying setpoint at shaft frequency (and harmonics) phase-locked to the reference (808) so that the support-axis component of the synchronous bearing load is reduced; the transverse component is not addressed by a support-force module and is left to conventional balancing. Both modes reduce measured bearing load without disassembly and are conditional on a qualified module class whose bandwidth covers the shaft frequency.

Verification. Bearing-load reduction is compared against passive balance-weight correction of equal mass; a phase scan verifies that the synchronous bearing load minimizes at one commanded phase and maximizes at the anti-phase, and a drive-off zero check separates the module's effect from sensor rectification.

PROPHETIC EXAMPLES (prospective; every number follows from the stated masses, constants, and assumed efficiencies; no result is represented as achieved)

P1 — cell. A 100 g YIG-class cell ($N_{fu} \approx 8.16\times10^{22}$; $E_{max} \approx 1.5$ J at 5.6 GHz by the 5 μ_B count, $\approx$ 1.1 J by $2M_s \cdot B \cdot V$ at room temperature) under 0.20 T bias would be charged at 50 W pulsed ($\dot{N} = 1.35\times10^{25}\ s^{-1}$; full-reversal ceiling reached in $\approx$ 30 ms of continuous drive; accessible energy at $\theta = 3° \approx$ 1.5 mJ) within coherence windows set by the measured T_1 (expected 0.3–1 μs at room temperature); state of charge would be read as line shift with temperature correction; the discharge coupler, decoupled during store, would connect an overcoupled pickup to a 50 Ω tuned load and rectifier; round-trip efficiency $\eta \approx \kappa_{ext}/(\kappa_{ext} + \kappa_{int}) \times$ (rectifier efficiency) is to be measured and reported, bounded below one. The long-storage protocol would hold for $t_h \geq 10 \cdot T_1$ and compare recovered energy with the τ_c–T_1 prediction under the bias-detuned, medium-removed, and gated controls.

P2 — plant. A 1000 t buffer over a 100 m head stores m·g·h = 9.81×10^8 J ≈ 272 kWh gross per stroke. With f = 0.3 on lift, the lift-stroke hoist force falls from 9.81 MN to 6.87 MN and the modules carry 2.94 MN for the lift; the shifted 30 % of lift work is drawn as off-peak module drive energy at the measured η_d. The ledger table (FIG. 8), with $\eta_{hoist} = \eta_{gen} = 0.9$ assumed and η_d symbolic until measured:

Quantity	**Formula**	**Untrimmed**	**f = 0.3, η_d = 1 (bound)**	**f = 0.3, η_d = 0.9**	**f = 0.3, η_d = 0.5**
Gross potential per stroke	m·g·h	272.4 kWh	272.4	272.4	272.4
Hoist electrical input	(1−f)·m·g·h/η_hoist	302.7 kWh	211.9	211.9	211.9
Module drive electrical input	f·m·g·h/η_d	0	81.7	90.8	163.4
Total off-peak draw	sum	302.7	293.6	302.7	375.3
Delivered on-peak (full-weight descent)	m·g·h·η_gen	245.2	245.2	245.2	245.2
Losses	inputs − delivered	57.5	48.4	57.5	130.1
Round-trip η	delivered/inputs	0.81	0.84	0.81	0.65
Lift-stroke hoist force	(1−f)·m·g	9.81 MN	6.87 MN	6.87 MN	6.87 MN
Module support-force change	f·m·g	0	2.94 MN	2.94 MN	2.94 MN
Generator braking load (option a)	m·g	9.81 MN	9.81 MN	9.81 MN	9.81 MN
Module mass fraction on buffer	m_mod/m	0	from class map	from class map	from class map

The η_d = 0.9 column is identical to the untrimmed plant, which is the receipt that trim alone does not shape rates; the η_d = 1 column is a bound, not a prediction. Round-trip efficiency is below one in every column.

P3 — isolation. A 500 kg bench on supports selected for full weight at fo = 2 Hz has k = m(2π·2 Hz)² = 7.90×10^4 N/m and static sag 4903 N / k = 6.2 cm. With f = 0.99 and supports re-selected to k′ = (1−f)·k = 790 N/m for the residual 49 N at the same 6.2 cm deflection, the suspension frequency would be 0.2 Hz; the modules would hold 4.85 kN continuously at the map's hold-energy characteristic. A 1 % trim-force error (49 N) would move the platform 6.2 cm —the full design deflection—so δF must be held below $k'\cdot x_m/S$ and the loop must add stiffness and damping; the stops (697) are rated for the full 4.9 kN. An intermediate f = 0.9375 would give 0.5 Hz. All values are to be measured.

P4 — bearing-load trim. A module class whose map covers a static offset equal to a stated fraction of a rotor assembly's weight would reduce the mean support-axis bearing load by that fraction; a class whose bandwidth covers a 50 Hz shaft frequency would reduce the support-axis synchronous component by an amount to be measured against the phase-scan verification. No magnitude is predicted.

POAMS-NATIVE STORAGE AND FORCE EMBODIMENTS

The conventional magnetization reversal, T1, T2*, and coupling-structure ring-down quantities are reference-species controls. They do not foreclose a distinct whole-turn winding channel, whose defining property is recovered energy or retained state beyond those controls. Likewise, reaction-frame sensors discriminate ordinary apparatus forces but do not supply or limit the POAMS constraint-bill mechanism. Direct prophetic storage and weight-trim embodiments remain part of the invention alongside empirically mapped production species.

CLAIMS

1. An energy storage cell comprising: a medium having a net uncompensated internal circulation and sufficient phase coherence under a substantially uniform static alignment field; an alignment-field source applying said alignment field; a resonant coupling structure; a charge coupler arranged to charge the medium's aligned population into a collective coherent precession state by pulsed excitation at a located precession resonance of the population; a discharge coupler arranged to extract energy inductively from relaxation of said state into a tuned load, and a decoupling element by which the discharge coupler is disconnected or detuned from the medium during storage; a cell calibration memory storing a measured energy-relaxation time of the medium, a stored-energy ceiling of the cell established from the medium's measured saturation magnetization, alignment field, and volume, and a validity envelope; and a controller configured to schedule charge pulses according to the measured energy-relaxation time, to inhibit charging beyond the stored-energy ceiling, to hold the alignment field and keep the discharge coupler decoupled during storage, and to connect the discharge coupler for discharge.
2. The cell of claim 1, further comprising a state-of-charge sensor comprising an absorption-spectroscopy probe of the medium's resonance line and a temperature witness, the controller correcting the probe reading by a stored line-shift-versus-temperature calibration and cross-checking it against metered recovered energy.
3. The cell of claim 1, operated at a cryogenic temperature at or below 77 K, wherein the cell calibration memory stores the energy-relaxation time measured at that temperature and the controller sets the storage interval from the stored value, the storage interval being extended only where the measured energy-relaxation time at that temperature exceeds the room-temperature value.
4. The cell of claim 1, wherein the medium is an insulating crystalline medium and the alignment field is at or above the medium's saturation value.
5. The cell of claim 1, further comprising a rectifier coupled to said tuned load, the cell thereby constituting a solid-state electrical power source.
6. The cell of claim 1, wherein the discharge coupler is overcoupled such that its external coupling rate to the tuned load exceeds the medium's intrinsic energy-relaxation rate.

7. The cell of claim 1, wherein the cell calibration memory further stores a long-storage status indicating whether a storage channel decaying more slowly than the measured energy-relaxation time has been identified for the cell's class by recovering energy after a hold interval exceeding the energy-relaxation time in excess of a prediction from the coupling-structure decay time and the energy-relaxation time under bias-detuned, medium-removed, and port-isolation controls with independent replication, and the controller extends the storage interval beyond the energy-relaxation time only when the long-storage status is set.
8. An energy storage system comprising a plurality of cells according to claim 1, a common charge/discharge bus, per-cell charge management enforcing each cell's stored-energy ceiling and thermal envelope, an array controller and power interface, and a common thermal plant.
9. A method of storing and recovering energy with the cell of claim 1, comprising: locating the precession resonance; charging the medium by pulsed resonant drive scheduled from the measured energy-relaxation time while metering absorbed energy and inhibiting charge beyond the stored-energy ceiling; storing with the alignment field held and the discharge coupler decoupled for an interval not exceeding the storage interval stored in the cell calibration memory; connecting the discharge coupler and discharging into the tuned load; and metering recovered energy and reporting a round-trip efficiency as a measured quantity less than one.
10. The method of claim 9, further comprising discriminating recovered energy of the population from energy stored in the coupling structure by repeating the charging and discharging with the alignment field detuned from resonance, gating the discharge measurement after drive-off with port isolation, and correcting the state-of-charge reading for temperature.
11. A method of supplying a load comprising charging the system of claim 8 during a first period and discharging into said load during a second period, the periods selected according to a rate or demand criterion, dispatch being limited to metered stored energy less measured losses.
12. A potential-energy storage plant comprising: a guide structure; a mass buffer movable along the guide structure between a lower level and an upper level; hoist machinery comprising a motor-generator coupled to the buffer; at least one weight-trim module mounted to the buffer, each weight-trim module comprising a body of ordered medium having an internal angular-momentum population alignable by a static field, an alignment-field source, a coupling structure and drive source configured to apply an oscillating drive to the body at a located precession resonance of the aligned population, a support interface transmitting a support force of the body rigidly to the buffer, a reaction frame carrying the alignment-field source and the coupling structure and transmitting their loads through a load path separate from the support interface, runtime witness sensors, a calibration memory storing an empirical support-force response map established by measurement of the module or of its production class and relating a drive state to a change in the support force of the body, the map containing at least one independently validated nonzero operating point, an uncertainty for each operating point, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and a module controller that

commands a drive state determined from the map for a support-force setpoint within the permissible interpolation region and otherwise inhibits the setpoint; energy and momentum metering configured to record hoist electrical input, module drive electrical input, generator output, cable tension, module support-path force, module reaction-path force, and guide-reaction force; and a scheduler configured to command, from the map, a trim setpoint equal to a fraction of the buffer's baseline support force during a lift stroke so that the lift-stroke hoist force is reduced by the measured achieved fraction, to limit scheduled dispatch to metered stored potential less measured losses, and to invalidate a trim record when the change in support force is accounted for within a stored tolerance by the module reaction-path force or the guide-reaction force.

13. The plant of claim 12, wherein the motor-generator is rated to brake a descent of the buffer at its full untrimmed weight, and the hoist machinery further comprises a lift motor rated for the trimmed lift-stroke force, or the motor-generator's rating is set by the full-weight descent and the trim setpoint reduces lift-stroke hoist force, cable tension, and motor current.
14. The plant of claim 12, wherein the scheduler is configured either to command descent at a speed reduced in proportion to the trimmed fraction with delivered power reduced in proportion, or to hold a trim setpoint during descent with the motor-generator rated for the trimmed descent load and the modules' descent-stroke drive energy entered in the ledger as an input.
15. The plant of claim 12, wherein the fraction commanded during the lift stroke is selected from at least one percent, at least ten percent, at least fifty percent, and substantially the whole of the buffer's baseline support force, each selection being permitted only where the permissible interpolation region of every operated module and a measured array allocation map of the plurality of modules cover the selected fraction.
16. The plant of claim 12, further comprising a lift-share actuator mounted to the buffer and reacting against the guide structure, the scheduler partitioning lift-stroke force among the hoist machinery, the lift-share actuator, and the weight-trim modules with the actuator's guide reaction metered.
17. The plant of claim 12, wherein the guide structure is a vertical shaft, a tower, or a slope rail, and cables, brakes, and buffer structure are rated for the full untrimmed weight.
18. A method of operating a potential-energy storage plant having a mass buffer, hoist machinery comprising a motor-generator, at least one weight-trim module mounted to the buffer and comprising a body of ordered medium, an alignment-field source, a coupling structure and drive source, a support interface to the buffer, a reaction frame on a separate load path, and a calibration memory storing an empirical support-force response map established by a qualification procedure performed before the method and containing at least one independently validated nonzero operating point, an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, the method comprising: verifying that the provenance record matches the module and that the module reports a valid state; commanding during a lift stroke a trim setpoint within the permissible interpolation region from the map while raising the buffer with the hoist machinery; metering hoist electrical input, module drive electrical input, cable tension,

module support-path force, module reaction-path force, and guide-reaction force during the lift stroke; transitioning the module to safe-off or holding the trim setpoint at the upper level according to a selected descent embodiment; lowering the buffer through the motor-generator during a generation stroke and metering generator output; maintaining an energy and momentum ledger in which recovered energy does not exceed module drive input plus hoist input less losses; limiting scheduled dispatch to metered stored potential less measured losses; and reporting round-trip efficiency as a measured quantity; wherein the qualification procedure is not repeated during the method.

19. The method of claim 18, further comprising recording in the ledger the module channel's measured energy characteristic as lift work transferred per unit module drive energy, and permitting a schedule that shifts grid draw or delivery in time relative to a plant without weight-trim modules only when the recorded characteristic supports it.

20. A vibration-isolation platform comprising: a platform for a payload; compliant supports between the platform and a base, the supports having a total stiffness selected for an untrimmed fraction of the platform's baseline weight at a design static deflection; support-path load cells in the compliant-support path; travel stops rated for the full platform weight; at least one weight-trim module whose support interface transmits a support force of a body of ordered medium rigidly to the platform, the module comprising an alignment-field source, a coupling structure and drive source configured to apply an oscillating drive to the body at a located precession resonance of its aligned internal-circulation population, a reaction frame on a load path separate from the support interface, runtime witness sensors, a support-path force sensor or connection to the load cells, a calibration memory storing an empirical support-force response map established by measurement and containing at least one independently validated nonzero operating point, an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and a module controller commanding a drive state from the map for a setpoint within the permissible interpolation region; and an isolation controller configured to command a trim setpoint equal to the trimmed fraction of the platform's baseline weight, to regulate the module on a trim error defined as the difference between the measured change in support force and the commanded trim force, and to transition the module to safe-off, returning the full platform weight to the compliant supports and travel stops, on a validity failure; whereby, with the supports' stiffness so selected, the platform's suspension natural frequency is approximately the square root of the untrimmed fraction times the natural frequency of supports selected for the full weight at the same deflection.

21. The platform of claim 20, wherein the trimmed fraction exceeds 0.9 and the isolation controller holds the trim-force error below the product of the residual support stiffness and a travel margin divided by a stated safety factor.

22. The platform of claim 21, wherein the trimmed fraction is substantially unity, the platform being substantially weight-null and free within the travel stops, and the isolation controller further comprises an inertial sensor on the platform providing velocity feedback for damping and stiffness

augmentation, the platform's suspension resonance being substantially eliminated within the map's bandwidth.

23. The platform of claim 20, wherein the calibration memory stores a measured position-dependent stiffness contribution of the module and the isolation controller includes it in the residual stiffness.

24. A method of operating a vibration-isolation platform having compliant supports selected for an untrimmed fraction of platform weight, travel stops rated for full weight, and at least one weight-trim module comprising a body of ordered medium, an alignment-field source, a coupling structure and drive source, a support interface to the platform, a reaction frame on a separate load path, and a calibration memory storing an empirical support-force response map established before the method and containing at least one independently validated nonzero operating point with uncertainty, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, the method comprising: measuring the platform's baseline weight with the module drive off; commanding a trim setpoint equal to the trimmed fraction of the baseline weight from the map within the permissible interpolation region; regulating the module on a trim error equal to the measured change in support force less the commanded trim force; verifying the residual support stiffness by static load-deflection; measuring the suspension natural frequency; and transitioning the module to safe-off onto the travel stops on a validity failure.

25. A machinery bearing-load trim system comprising: a machine having a rotating or reciprocating element in bearings; bearing-load sensors; a once-per-revolution phase reference; at least one weight-trim module mounted rigidly to a bearing housing, pedestal, or machine element so that a change in the support force of the module's body appears in the bearing load, the module comprising a body of ordered medium, an alignment-field source, a coupling structure and drive source configured to apply an oscillating drive to the body at a located precession resonance of its aligned internal-circulation population, a reaction frame on a load path separate from the module's support interface, runtime witness sensors, a calibration memory storing an empirical support-force response map established by measurement in the module's mounting class and containing at least one independently validated nonzero operating point, an uncertainty, a bandwidth, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and a module controller commanding a drive state from the map for a setpoint within the permissible interpolation region; and a trim controller configured to command a static trim setpoint reducing the mean support-axis bearing load by a commanded fraction, and, where the map's bandwidth covers the shaft frequency, a time-varying setpoint at the shaft frequency phase-locked to the phase reference so that the support-axis synchronous component of the bearing load is reduced, the reduction being reported from the bearing-load sensors as a measured quantity.

26. The system of claim 25, wherein the trim controller performs a phase scan of the time-varying setpoint and accepts the synchronous offset only when the measured synchronous bearing load is minimized at one commanded phase and maximized at the opposite phase and a drive-off zero check reproduces the untrimmed load.

27. A method of trimming bearing load of machinery in situ, comprising: mounting at least one weight-trim module comprising a body of ordered medium, an alignment-field source, a coupling structure and drive source, a support interface, a reaction frame on a separate load path, and a calibration memory storing an empirical support-force response map established before the method and containing at least one independently validated nonzero operating point with uncertainty, bandwidth, provenance, a permissible interpolation region, runtime limits, and safe-off parameters, to a bearing housing, pedestal, or machine element; measuring bearing load with the module drive off; commanding from the map a static trim setpoint within the permissible interpolation region and measuring the change in mean bearing load; commanding, where the map's bandwidth covers the shaft frequency, a time-varying setpoint phase-locked to a once-per-revolution reference and scanning its phase; and reporting the measured reduction in support-axis bearing load.
28. A prophetic POAMS-native energy storage cell according to claim 1, wherein completed whole turns of the driven angular-momentum population constitute a winding channel distinct from the conventional magnetization-reversal channel, and the controller writes, holds, and discharges that channel beyond the reference-species bound or relaxation time when the channel is identified, without treating 2M_sBV, T1, T2*, or cavity ring-down as a universal ceiling on the winding state.
29. A plant, platform, or bearing-load system according to any one of claims 12, 20, and 25, wherein the weight-trim module operates prophetically by reallocating angular momentum and changing the supported body's natural-orbit-radius constraint bill, the reaction-frame measurement serving as an artifact control and not as a required reaction partner for the POAMS-native change.

ABSTRACT

Energy and infrastructure products built on coherently driven angular-momentum populations. A solid-state coherence storage cell charges an aligned medium into a collective coherent precession state by pulsed resonant drive, holds it with the discharge coupler decoupled, and discharges inductively through an overcoupled coupler to a tuned load and rectifier; a cell calibration memory stores the measured energy-relaxation time, the stored-energy ceiling $E_{max} \approx 2M_s \cdot B \cdot V$, and a temperature-corrected state-of-charge calibration; arrays and time-of-use dispatch are disclosed, with storage beyond the energy-relaxation time an identified hypothesis under a stated gate. A weight-trim module—ordered medium, alignment source, resonant drive, support interface, separate reaction path, and an empirically established support-force response map—is reproduced inline and applied to a potential-energy plant with explicit machine sizing and an energy-and-momentum ledger, to an isolation platform whose supports are selected for the untrimmed weight, and to in-situ bearing-load trim. No over-unity operation; all results to be measured.

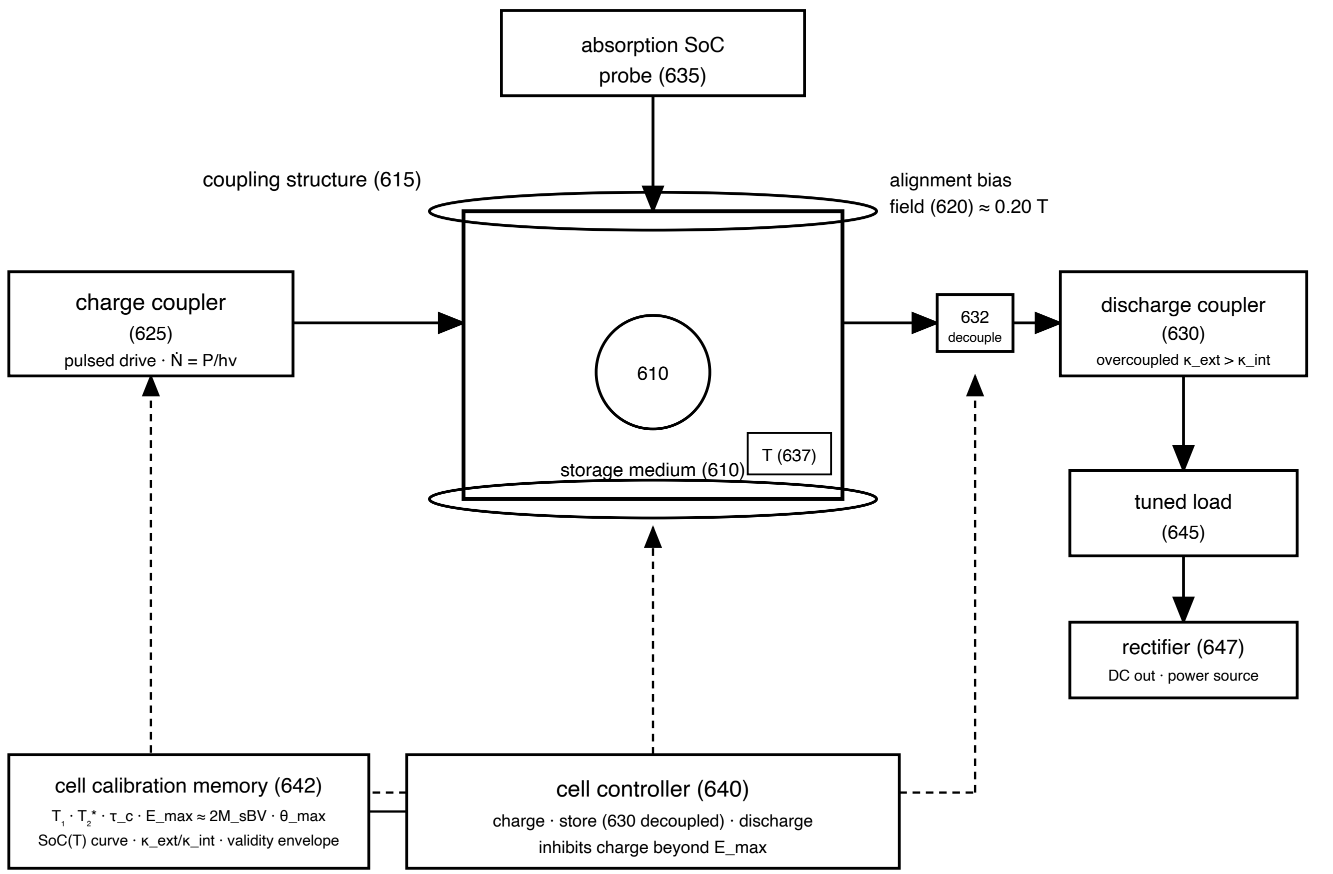

FIG. 1

Coherence storage cell (600) — medium in coupling structure under alignment bias, charge coupler, decoupling element, overcoupled discharge coupler, SoC probe, temperature witness, controller, cell calibration memory, tuned load, rectifier.

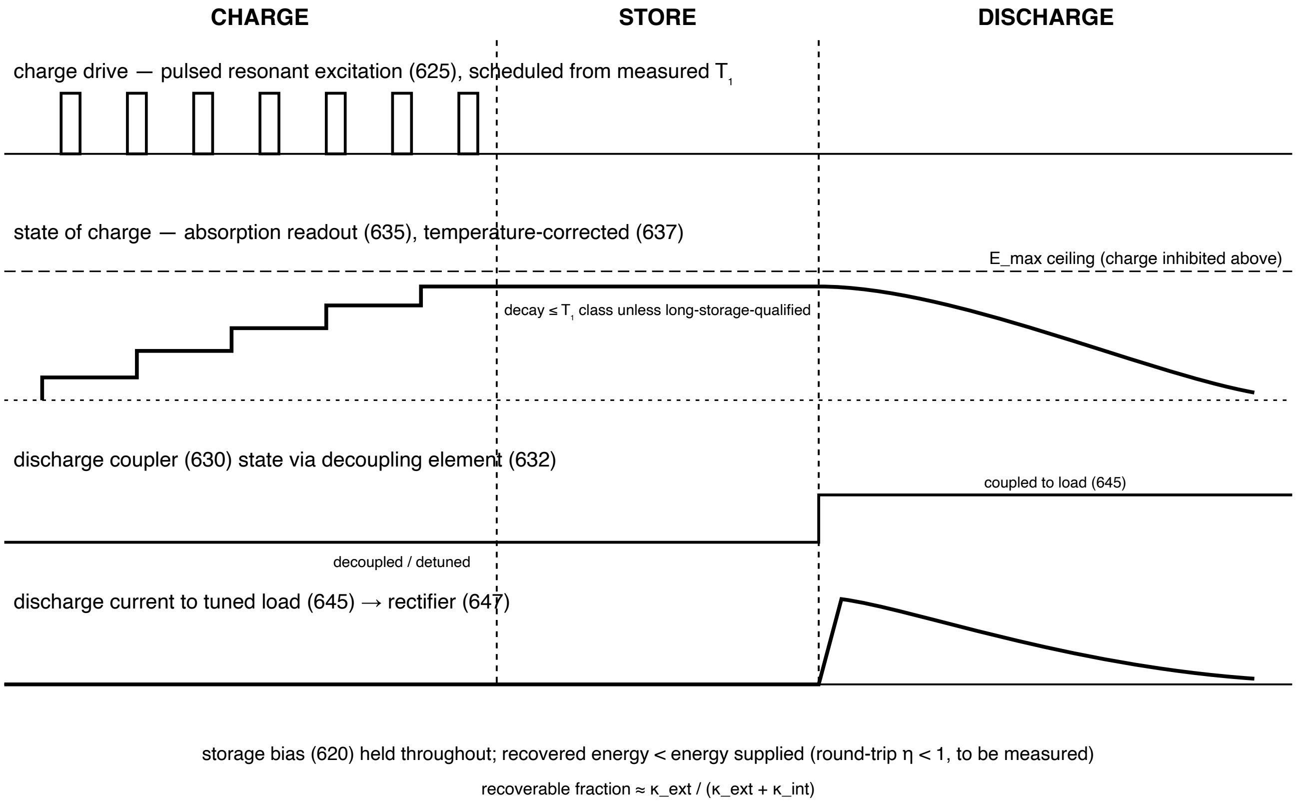

FIG. 2

Cycle timing — charge pulse train below the stored ceiling, absorption state-of-charge curve, discharge coupler decoupled during store, inductive discharge current.

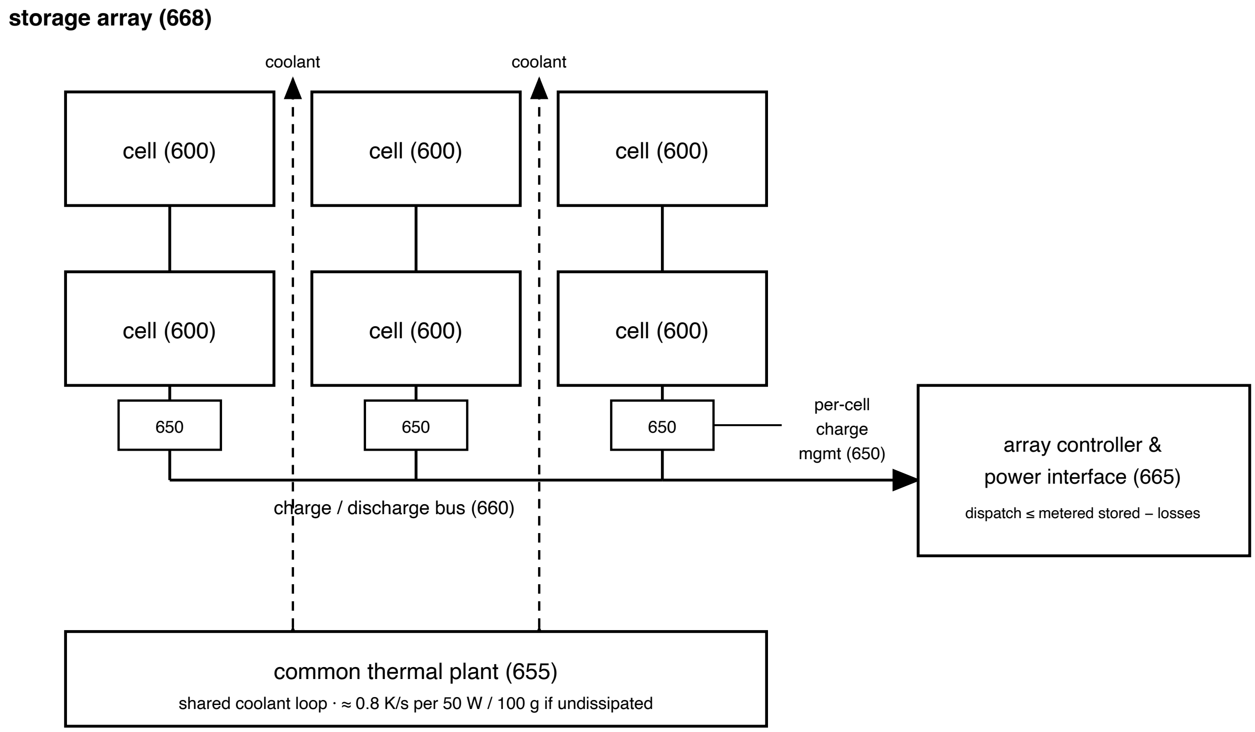

FIG. 3

Storage array (668) — cells on a common bus with per-cell charge management, array controller and power interface, and shared thermal plant.

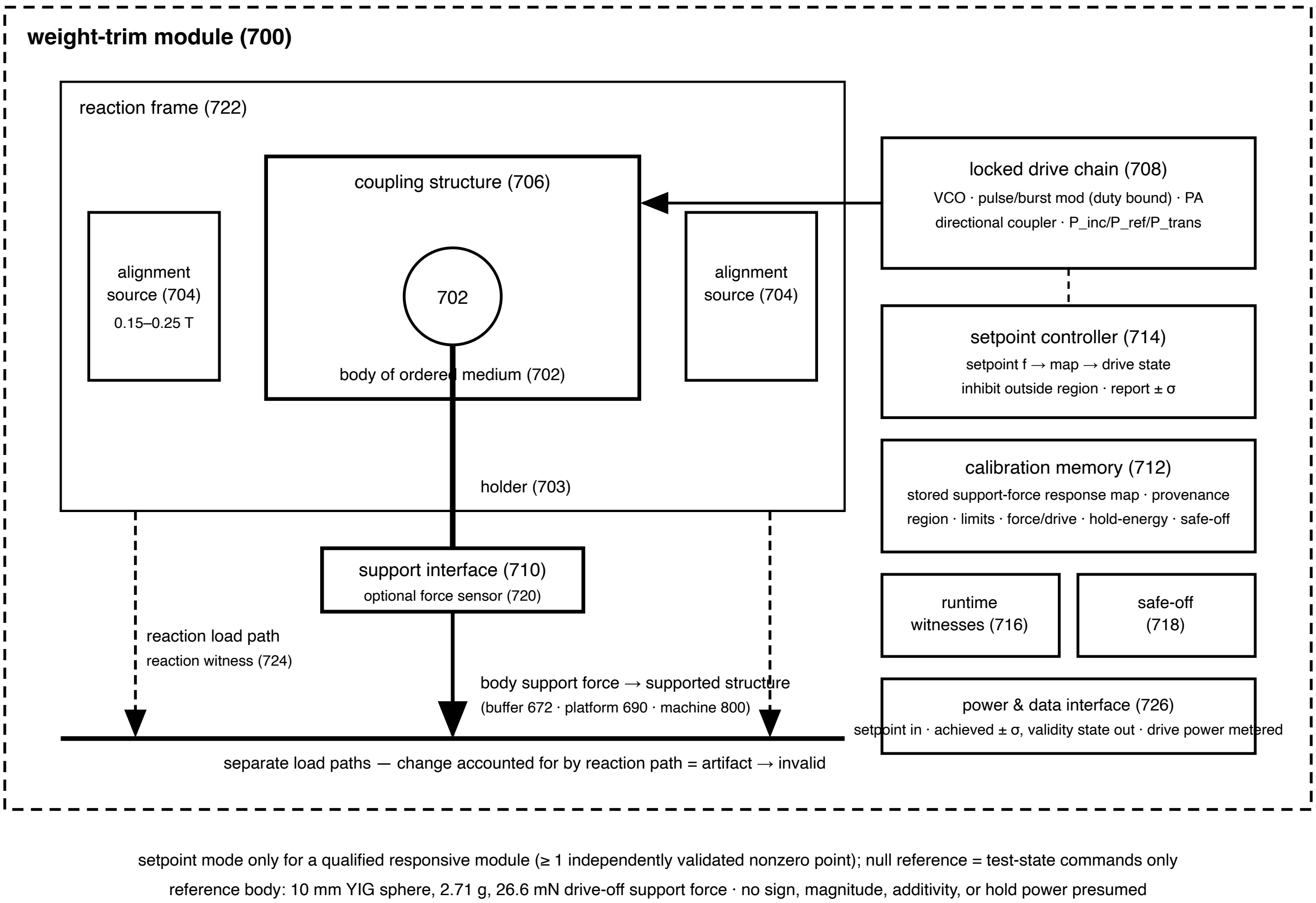

FIG. 4

Weight-trim module (700) — common species for D2–D4: ordered medium in coupling structure on a separately loaded reaction frame, support interface, calibration memory with stored response map, setpoint controller, witnesses, safe-off, interface.

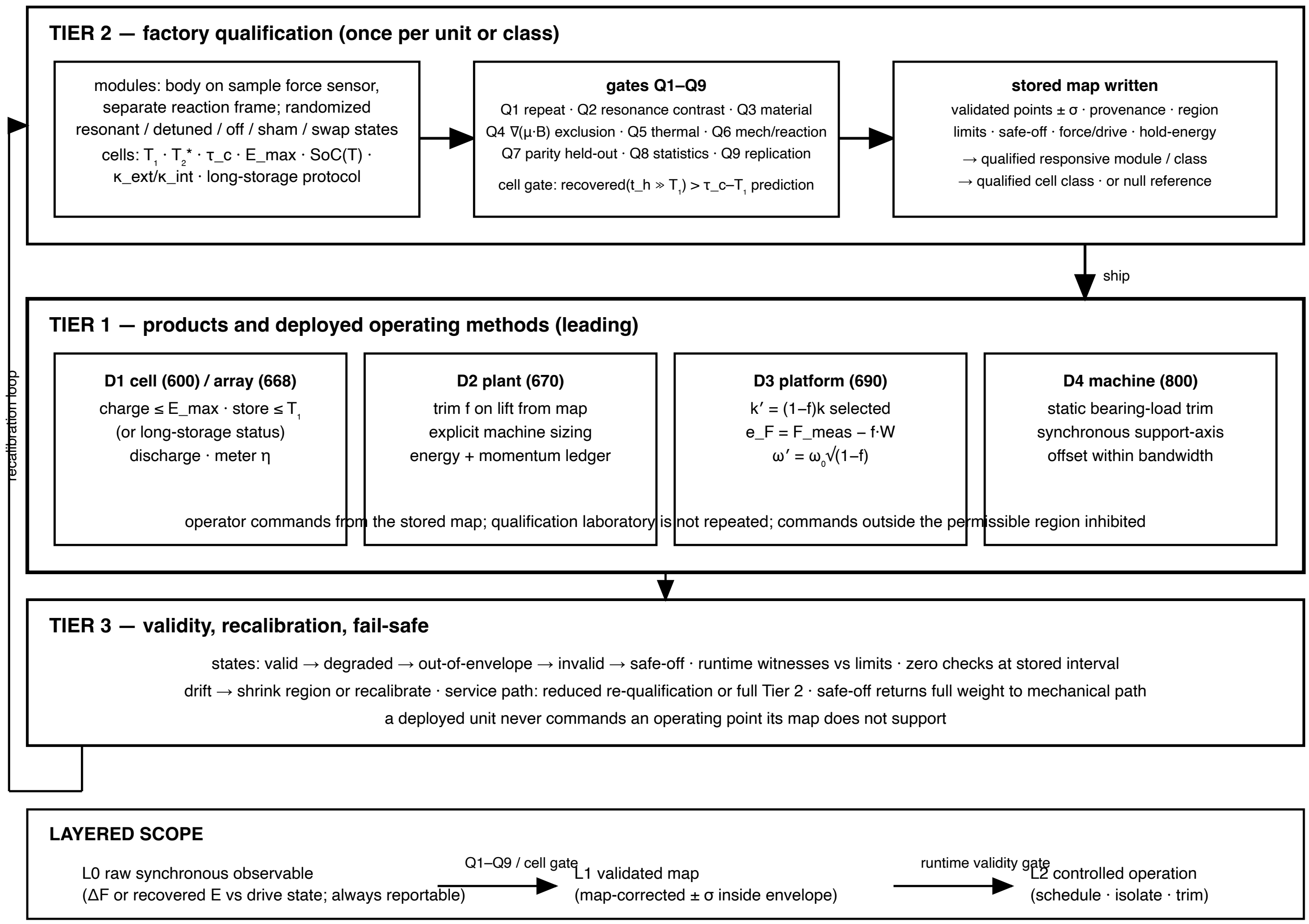

FIG. 5

Three-tier architecture and layered scope — factory qualification with gates Q1–Q9 produces the stored map; products operate from it; validity states and recalibration govern deployed commands; L0 → L1 → L2.

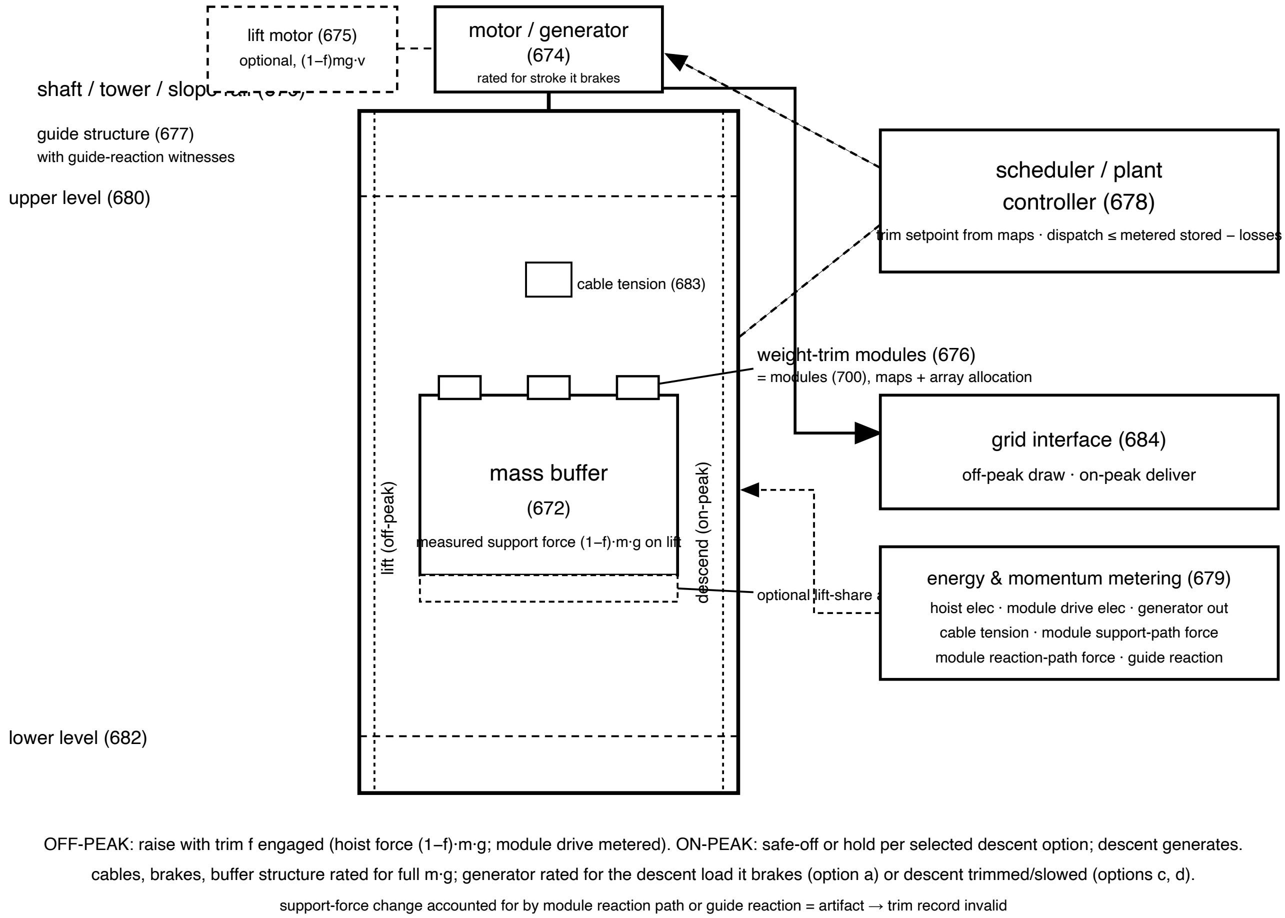

FIG. 6

Weight-trimmed potential-energy plant (670) — shaft with instrumented guide structure, mass buffer, motor-generator and optional lift motor, weight-trim modules, optional lift-share actuator, scheduler, metering, grid interface.

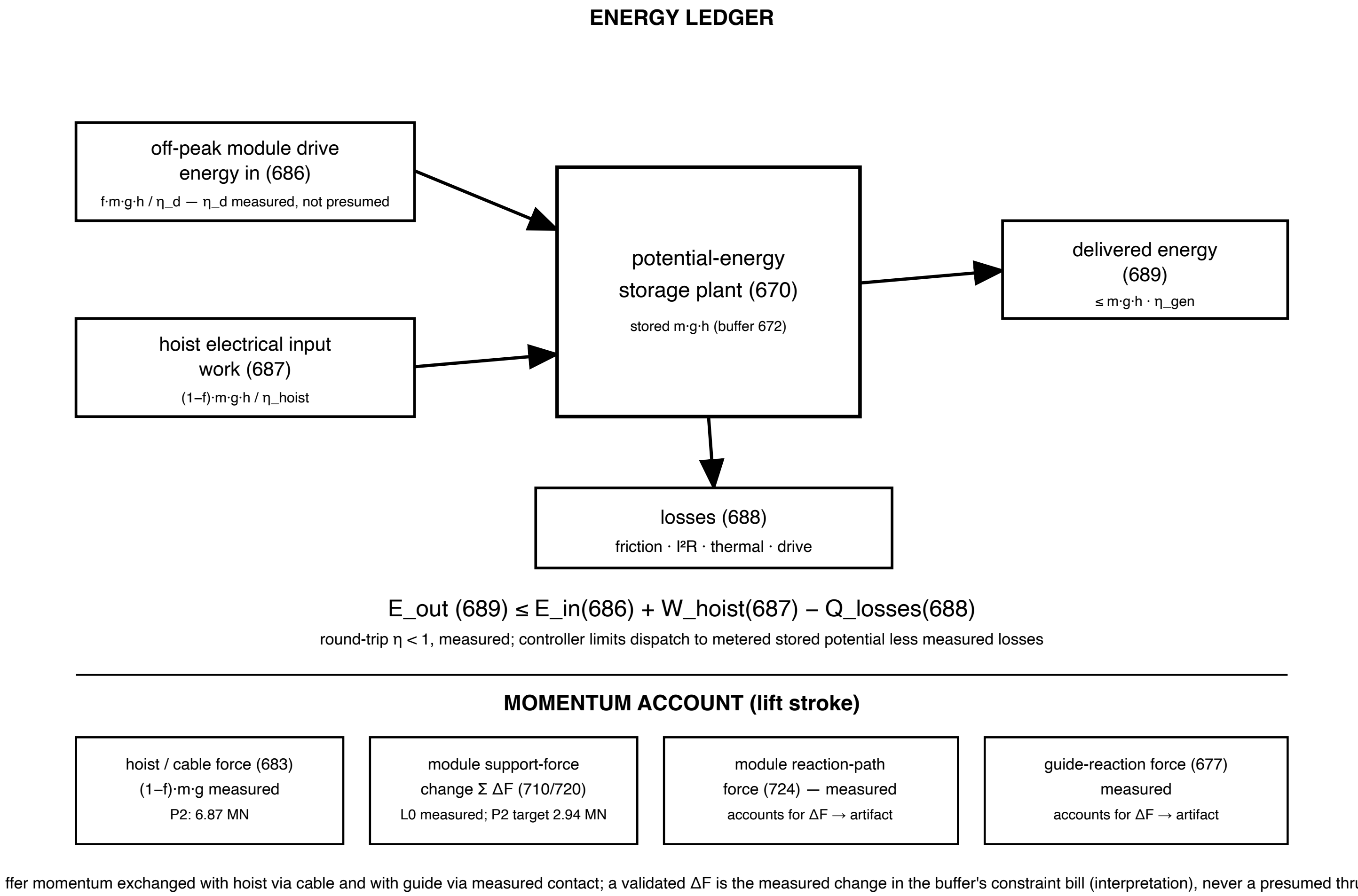

FIG. 7

Plant energy and momentum ledger — module drive and hoist electrical inputs, delivered output, losses, conservation bound; measured hoist, module, reaction-path, and guide forces.

LEDGER TABLE — P2 sizing example: $m = 1000$ t, $h = 100$ m, $f = 0.3$; $\eta_{hoist} = \eta_{gen} = 0.9$ assumed; η_d to be measured

Quantity	Formula	Untrimmed	$\eta_d = 1$ (bound)	$\eta_d = 0.9$	$\eta_d = 0.5$
Gross potential per stroke	$m \cdot g \cdot h$	272.4 kWh	272.4	272.4	272.4
Hoist electrical input	$(1-f) \cdot m \cdot g \cdot h / \eta_{hoist}$	302.7	211.9	211.9	211.9
Module drive electrical input	$f \cdot m \cdot g \cdot h / \eta_d$	0	81.7	90.8	163.4
Total off-peak draw	sum	302.7	293.6	302.7	375.3
Delivered on-peak (full-weight descent)	$m \cdot g \cdot h \cdot \eta_{gen}$	245.2	245.2	245.2	245.2
Losses	inputs – delivered	57.5	48.4	57.5	130.1
Round-trip η	delivered/inputs	0.81	0.84	0.81	0.65
Lift-stroke hoist force	$(1-f) \cdot m \cdot g$	9.81 MN	6.87 MN	6.87 MN	6.87 MN
Module support-force change	$f \cdot m \cdot g$	0	2.94 MN	2.94 MN	2.94 MN
Generator braking load (option a)	$m \cdot g$	9.81 MN	9.81 MN	9.81 MN	9.81 MN
Module mass fraction on buffer	m_{mod}/m	0	from class map	from class map	from class map

$\eta_d = 0.9$ column ≡ untrimmed plant: trim alone does not shape rates. $\eta_d = 1$ column is a bound, not a prediction. Round-trip $\eta < 1$ in every column.

$g = 9.80665$ m/s²; $m \cdot g \cdot h = 9.807 \times 10^8$ J. Rate shaping beyond a conventional plant only if measured $\eta_d > \eta_{hoist}$ or module draw is time-decoupled — not presumed.

Machine-sizing options

(a) generator rated $m \cdot g \cdot v_{descent}$, lift channel $(1-f) \cdot m \cdot g \cdot v_{lift}$; single machine nameplate set by descent · (b) separate lift motor (675) + generator (674)

(c) descent at $(1-f) \cdot v_{lift}$, delivered power reduced in proportion · (d) trim f' held on descent, machine rated $(1-f') \cdot m \cdot g \cdot v$, module descent drive energy = cost unless map records otherwise

cables, brakes, buffer structure rated for full $m \cdot g$ in every option (safe-off returns full weight to hoist path)

FIG. 8

Ledger table for the P2 sizing example — energy and force rows, untrimmed reference and module-efficiency columns, and the four machine-sizing options.

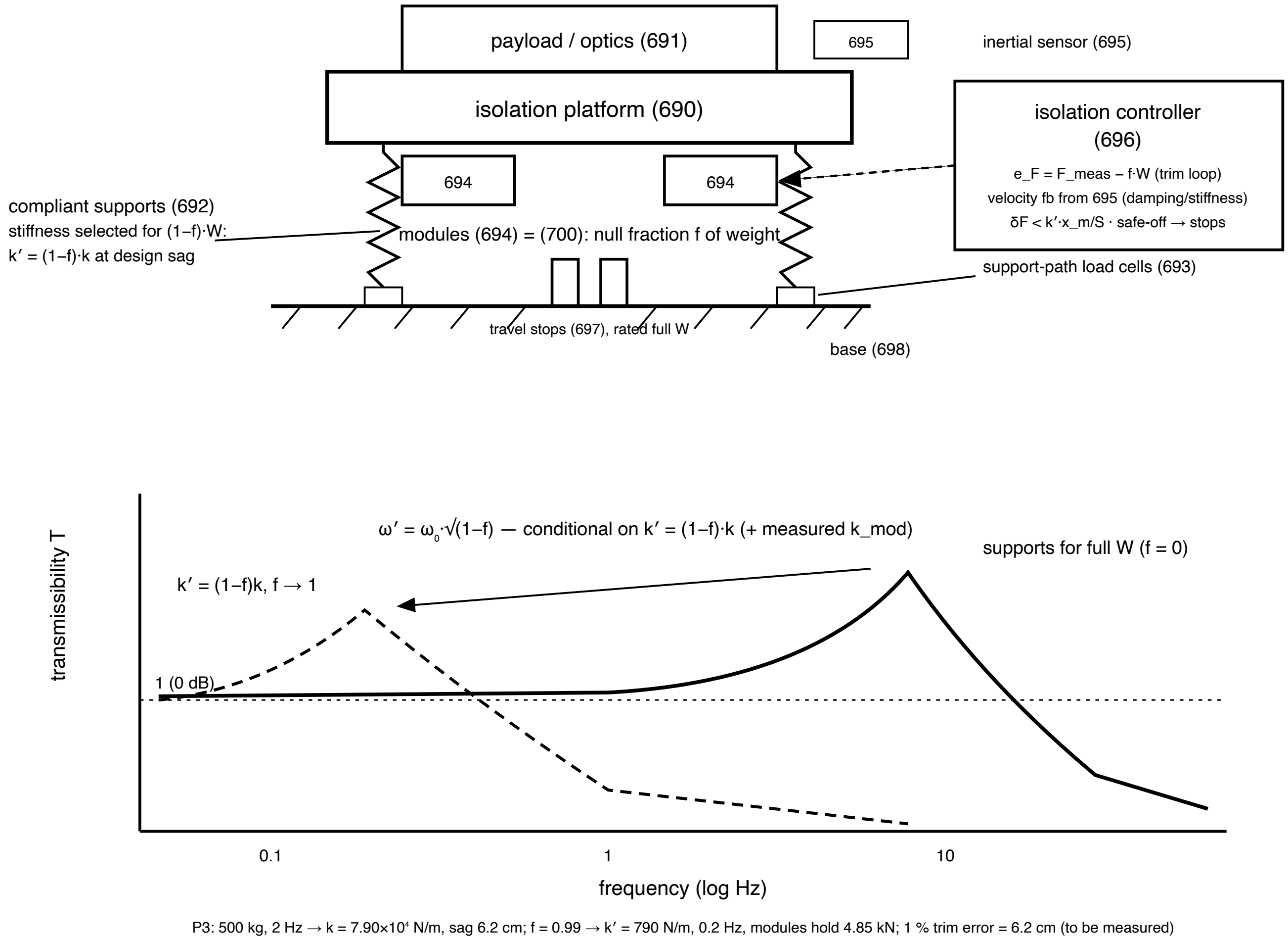

FIG. 9

Weight-trim isolation platform (690) and transmissibility — supports selected for the untrimmed fraction, defined error signal, full-weight stops; resonance lowered as √(1−f) under the stiffness-selection condition.

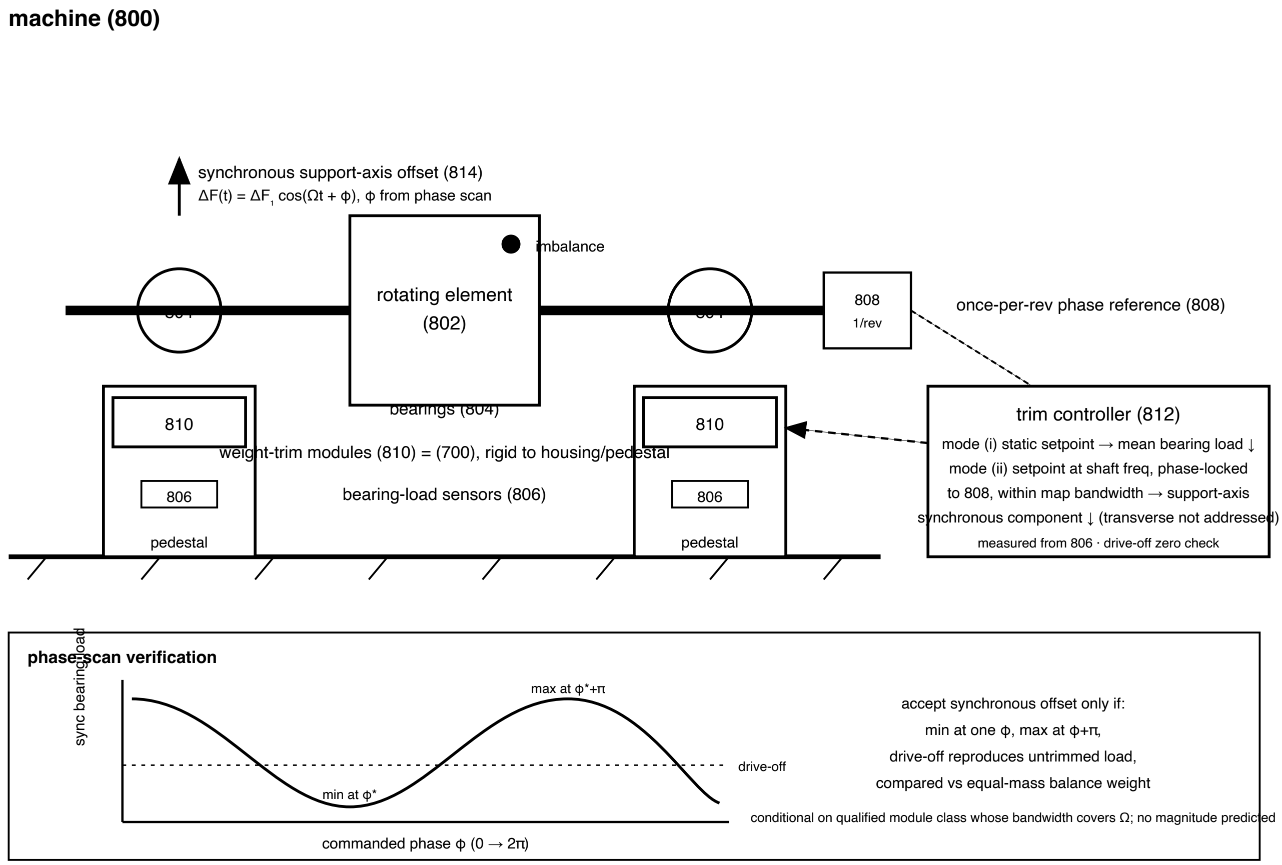

FIG. 10

Machinery bearing-load trim system (800) — modules on bearing pedestals, bearing-load sensors, once-per-rev reference, trim controller with static and shaft-synchronous modes; phase-scan verification inset.